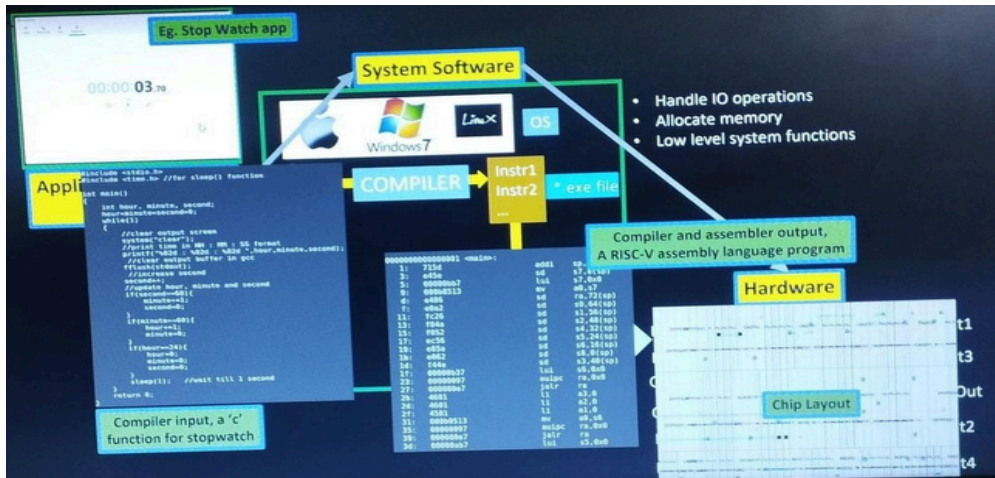


CHIP DESIGNING COURSE

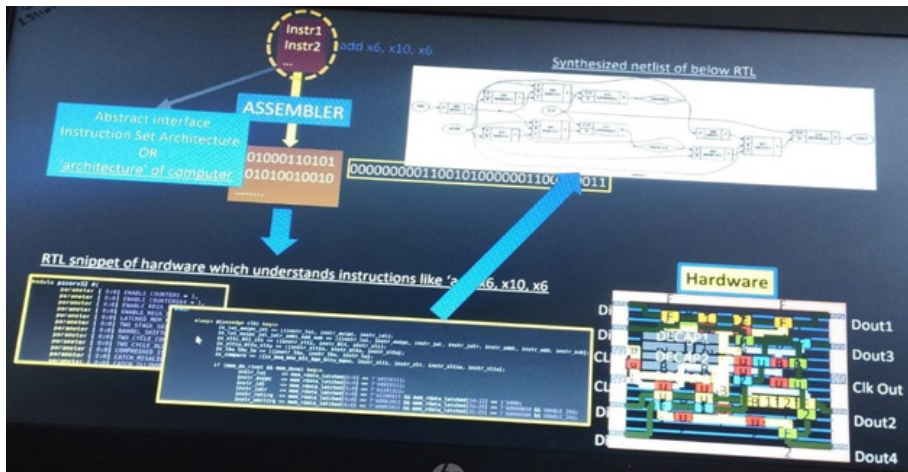
ASIC AND OPEN ASIC FLOWS

Stop watch



In the course we will be concentrating on RISC V architecture.

Eg.

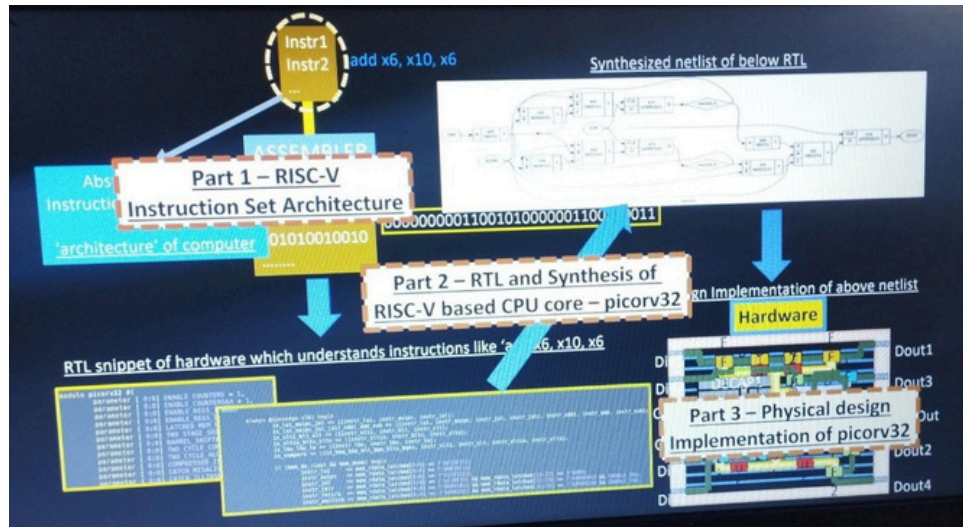


For the command to add x6,x10,x6 ...the assembler result and the binary out put are given .Hardware description language -Which in turn RTL is synthesized in netlist as in the above pic.

Then Physical design implementation of the above netlist happens.

THE COURSE AS PARTS

CHIP DESIGNING COURSE

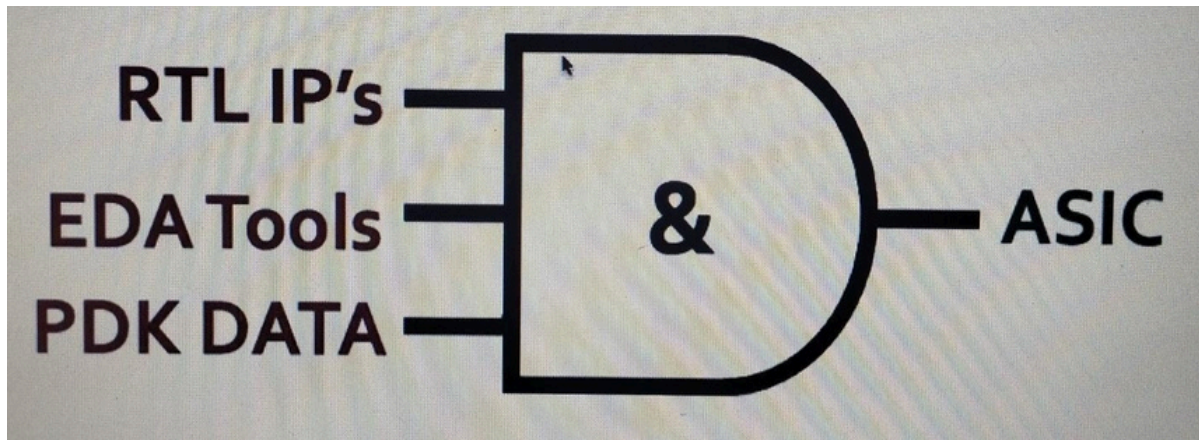


Part 1A

INSTRUCTION SET ARCHITECTURE

Digital ASIC design

ASIC - Application specific integrated circuit



PDK - process Design kit. Till 1979 IC design and the application was practiced by few companies like TI and INTEL. In 1979 Convey and carver advocated the need for separating the design from the technology. Since then we started to see pure play (only designing) Fabless companies and fabrication only companies

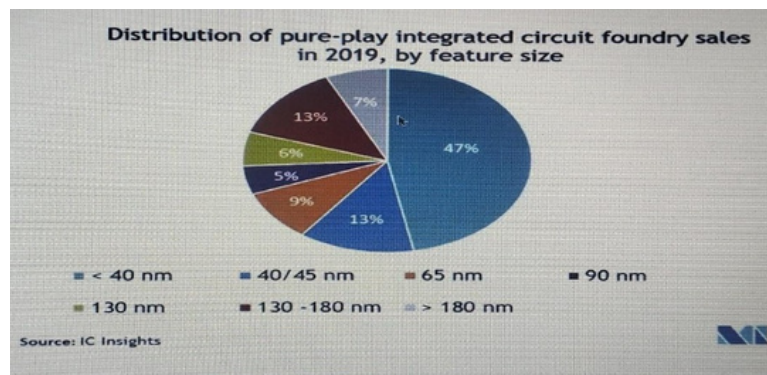
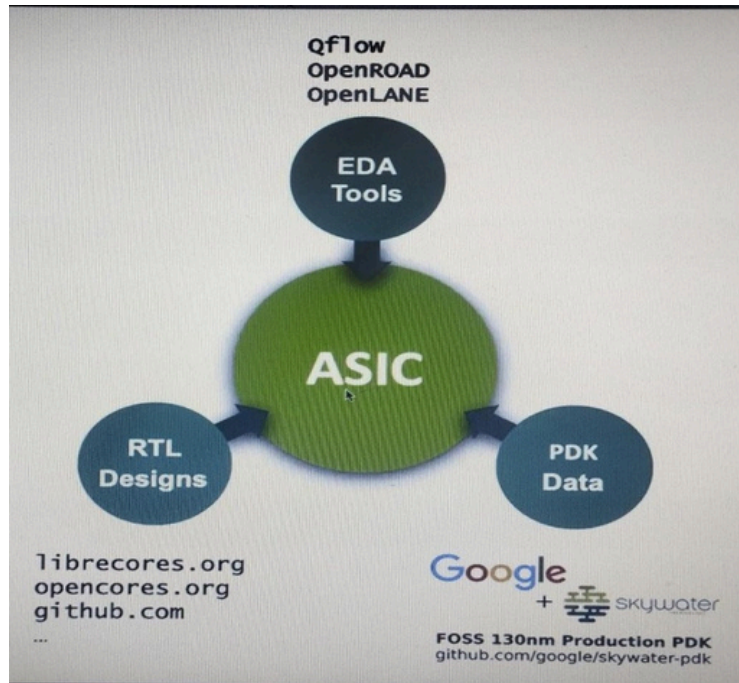
CHIP DESIGNING COURSE

Interface between the designers and the fab become a set of data files referred to PDK files.

PDK contains collection of files used to model a fabrication process for the EDA tools used to design an IC

- Process design rules:DRC,LVS.PEX
- DEVICE MODELS
- DIGITAL STANDARD CELL LIBRARIS
- I/O libraries

Under non-disclosure agreements which made is difficult for enable IC fabrication for masses...this is not the case anymore as in 2020 GOOGLE + SKYWATER agreed to open source PDK for 130nm .As a result in june 2020 google released the 1st ever opensource PDK for the masses.Now we got all 3 elemets for open source ASIC.



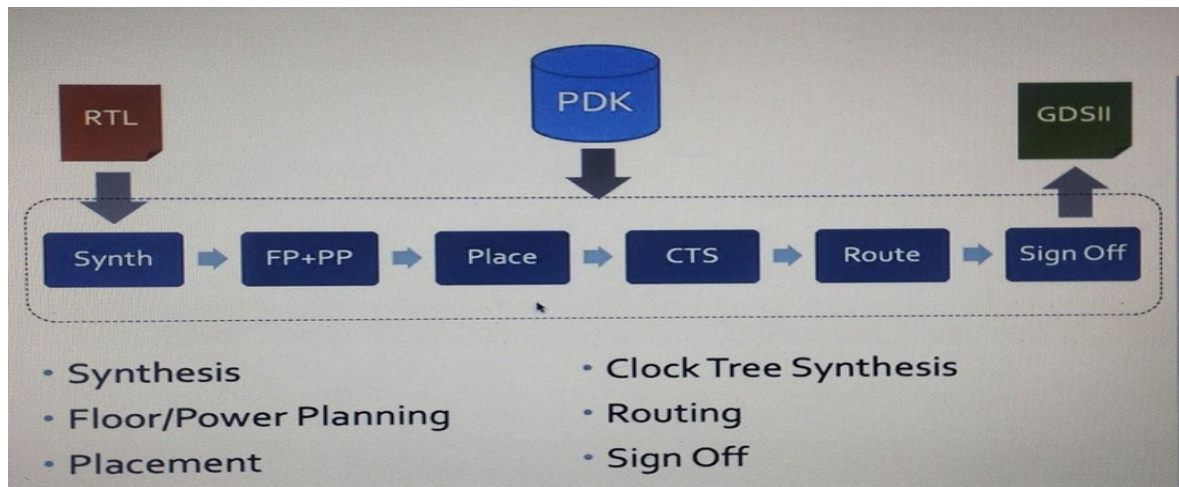
CHIP DESIGNING COURSE

Titbit : IBM has produced the smallest chip at 2 nm

The fabrication cost of 130nm is cheaper.

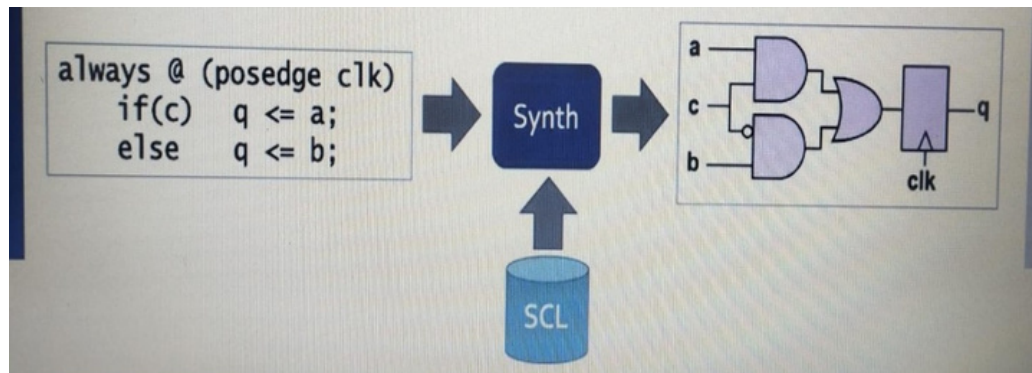
ASIC flow objective : **RTL to GDSII** (also called Automated PnR OR Physical implementation)

Simplified RTL- GDSii



SYNTHESIS

Converts RTL to a circuit out of components from the standard cell library (SCL).Resultant circuit will be in HDL.gate level netlist.



Standard cells have regular layout

Each has different views/model

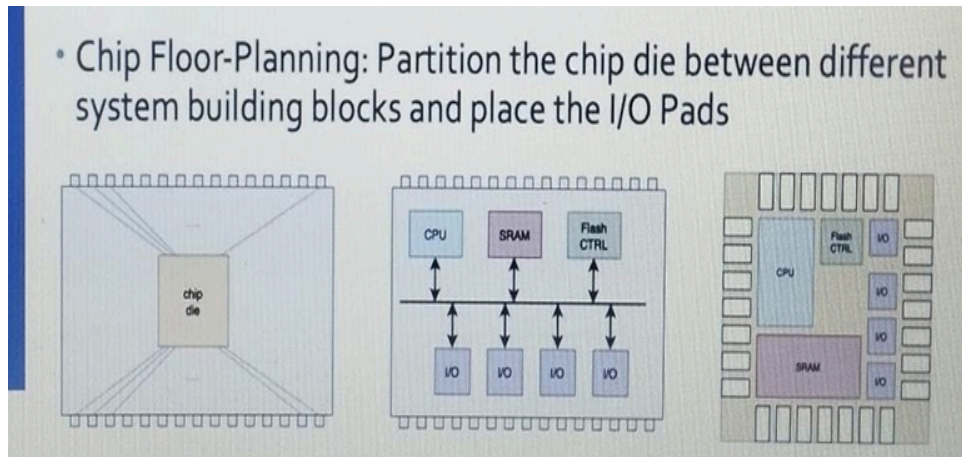
- Electrical,HDL,SPICE
- LAYOUT

Floor planning

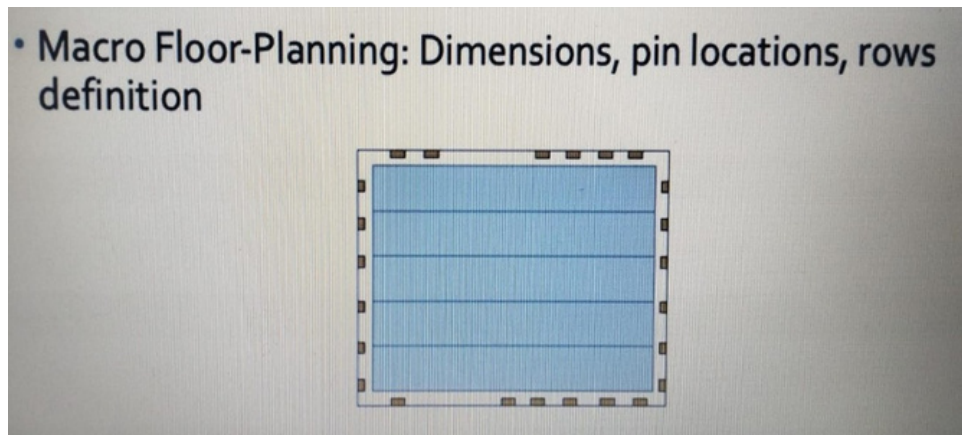
Can be for a chip or for the package as a whole

CHIP DESIGNING COURSE

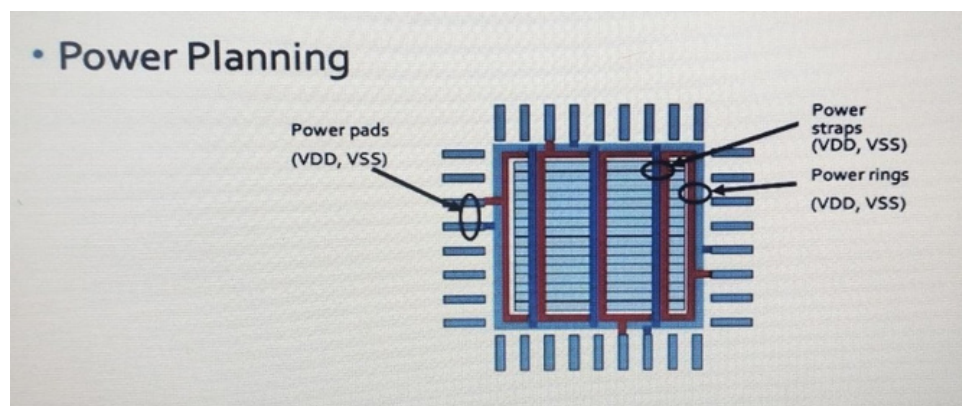
Chip floor planning



Macro floor planning



Power planning

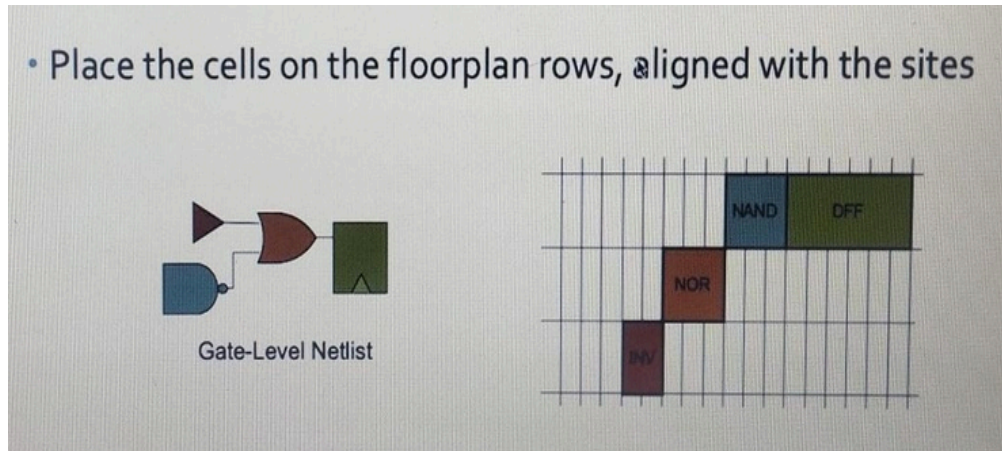


It is powered by multiple power pads (VDD AND VSS).Power pads are connected to all components in horizontal and vertical metal pads. Such parallel structures are meant to reduce the resistance.

CHIP DESIGNING COURSE

To address the electro magnetic problem, typically it uses the upper metal layer as they are thicker than lower metal layer hence have less resistance.

Placement

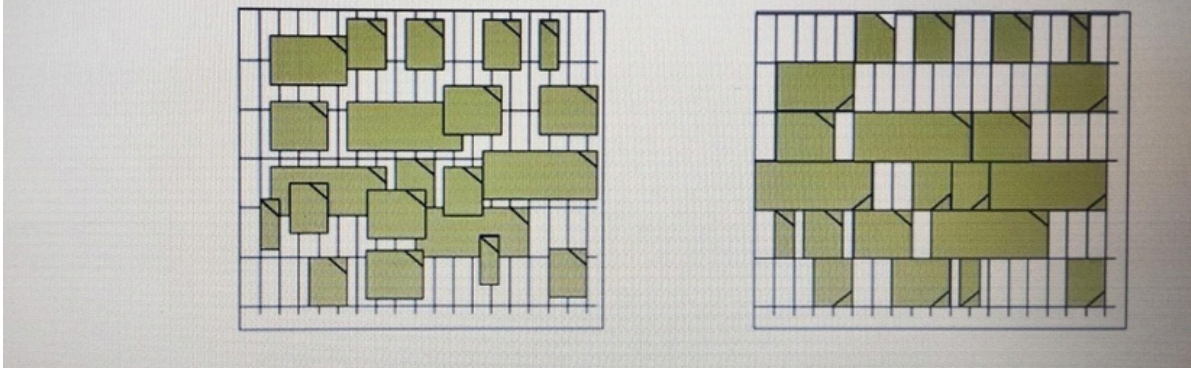


Macros would Place the gate level net list results on the virtual rows and the cells will be placed close to each other to avoid the interconnect delay and for successful routing. Typically cell placements are done in 2 ways .

Global position : Optimal placement for all cells such positions are not necessarily legal .cells may overlap.

Detailed placement : Minimally altered to be legal

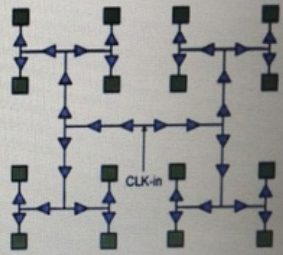
- Usually done in 2 steps: Global and Detailed



CLOCK TREE SYNTHESIS

Before routing the signals we need to route the clock. The clock network looks like a tree. Clock source is the root and the clock elements are the leaves. Clock size is at the level of the tree.

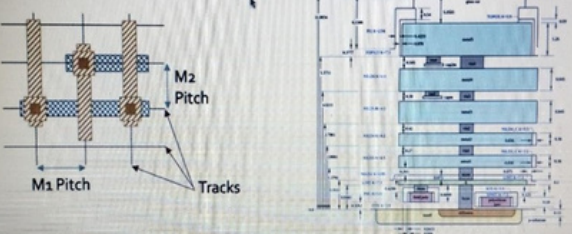
- Create a clock distribution network
 - To deliver the clock to all sequential elements (e.g., FF)
 - With minimum skew (zero is hard to achieve)
 - And in a good shape
 - Usually a Tree (H, X, ...)



ROUTING

Implementing the interconnect using the available metal layers –patterns .Router uses the available metal layers as defined by the PDK.

- Implement the interconnect using the available metal layers



For each metal layer the PDK defines the pitch, thickness, the minimum width and the VS.

SKY130 PDK 6 routing layers lowest layer is called the local interconnect layer is **titanium** ,following 5 layers local interconnect layers are all **aluminium** layers
Metal tracks form a routing grid

Routing grid is huge

Divide and conquer

- global routing generates routing guides –coarse grains
- detailed routing uses the routing guides to implement the actual wiring – fine grain

SIGNOFF

After routing final layout

Physical verifications

CHIP DESIGNING COURSE

- Design rules checking(DRC)
- Layout vs Schematic (LVS) - Final layout matches the gatelevel network

Timing verification

- Static timing analysis (STA) all timings matches

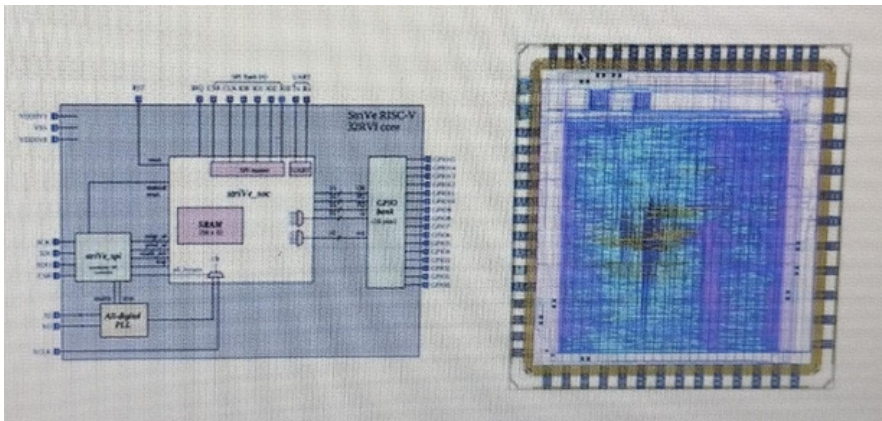
The problem is tougher when using open source EDA

- Tools qualification
- Tools calibration
- Missing tools

To ensure there is qualified tools , calibrated and no tool is missing it was decided to create open source ASIC methodology of flow calling it OPENLANE.

OPENLANE

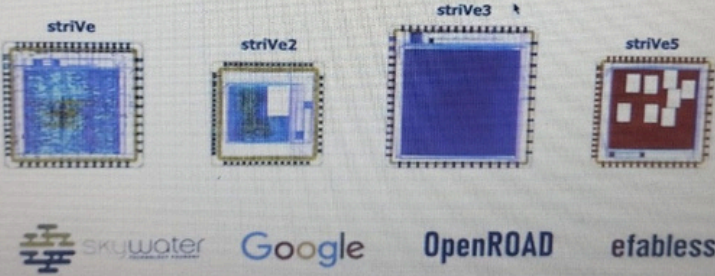
Started as a Open source flow for a true open source tape out experiment.



striVe is a family of open everything SoCs -open PDK,open EDA,open RTL.

CHIP DESIGNING COURSE

SoC	Features
<i>striVe</i>	Sky130 SCL + Synthesized 1 Kbytes SRAM
<i>striVe 2</i>	Sky130 SCL + 1 Kbytes OpenRAM block
<i>striVe 2a</i>	<i>striVe 2</i> with a single chip core module
<i>striVe 3</i>	OSU SCL + Synthesized 1 Kbytes SRAM
<i>striVe 5</i>	Sky130 SCL + 8 x 1 Kbytes OpenRAM banks
<i>striVe 6</i>	<i>striVe 2</i> with DFT



Main goal of OPEN LANE ASIC FLOW

Produce a clean GDSII with no human intervention .

Clean means

- No LVS violations
- No DRC violations
- Timing violations (work in progress)

Tuned for FOR SKYWATER 130nm Open PDK

Containerized

- Functional out of the box
- Instructions to build and run natively will follow

Can be used to harden Macros and chips

Two modes of operation Autonomous and Interactive

- Autonomus is push button flow
- Interactive commands and steps can be run one by one.

Design space exploration

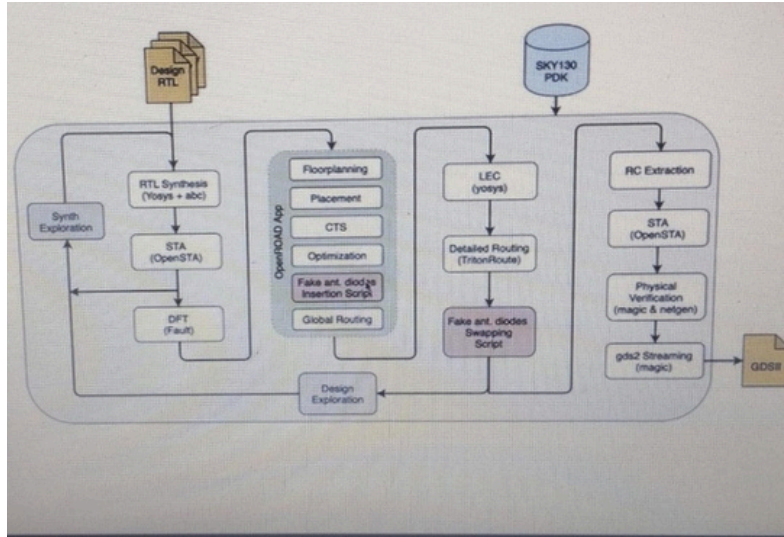
- Find the best set of flow configurations

Large number of design examples

- 43 designs with their best configurations
- More to be assed soon

OPENLANE ASIC FLOW

CHIP DESIGNING COURSE

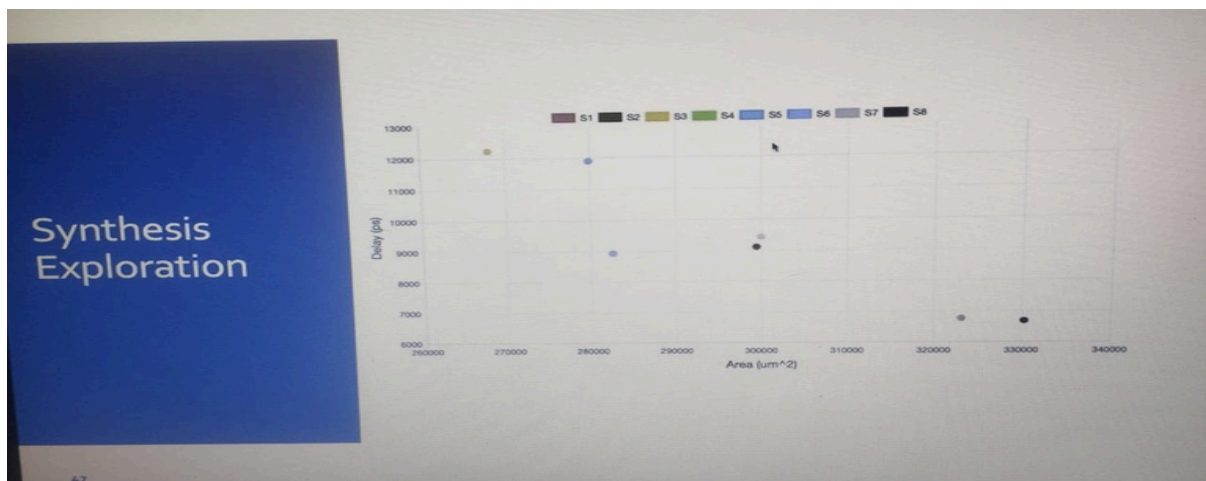


The flow starts with the design of RTL and ends with GDSII (final layout) to function it needs PDK.

OPENLANE is based on several opensource pocchets like openroad,klayout,magic,fault,yosys,qflow and ABC.

The flow starts with RTL synthesis -(RTL is fed to yosys with the design constains , yosys translates the RTL into logic store kits using components - these can be optimized using cells componants and ABC) ABC has to be guided during optimization.

SYNTHESIS STRATERGIES - STAdifferent designs can use different strategies to achieve the objectives for which we have synthesis exploration is available .Synthesis exploration utility used to generate that shows the design is affected and helps to find the best strategy to continue with.



Design exploration utility

M.SRIGHUKAN , 7th THE BVB ,ERODE

CHIP DESIGNING COURSE

Which helps to sweep design configurations which is more than 16 and generates report as below

aes	1h29m8s	22932	1	40	1	0.05
aes	1h34m31s	22932	2	30	1	0.05
aes	1h41m14s	22932	9	40	1	0.05
aes	1h47m14s	22932	1	45	1	0.05
aes	1h44m14s	22932	1	40	1	0.05
aes	1h47m59s	22932	1	45	1	0.05
aes	1h49m7s	22932	1	45	1	0.05
aes	1h43m54s	22932	2	30	1	0.05
aes	1h42m58s	22932	8	30	1	0.05
cordic	0h10m51s	8275	0	45	0	0.15
cordic	0h10m35s	8275	0	45	0	0.15
cordic	0h9m55s	8275	2	40	0	0.15
cordic	0h11m25s	8275	0	45	0	0.15
cordic	0h10m3s	8275	0	30	0	0.15
cordic	0h11m6s	8275	4	40	0	0.15
cordic	0h11m3s	8275	4	40	0	0.15
cordic	0h10m25s	8275	0	30	0	0.15
cordic	0h10m26s	8275	3	30	0	0.15

Very useful in finding the configurations for openlane.so its recommended to explore the design 1st and use the best configuration.

Design exploration utility can also be used for regression testing (ci)

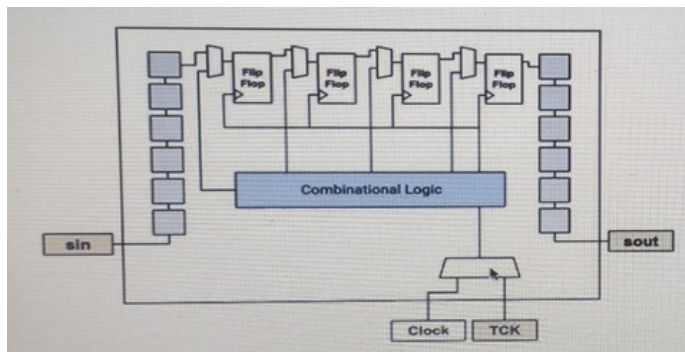
We run openlane on 70 designs and compare the results to the best ones

CHIP DESIGNING COURSE

Design	Runtime	Cell Count	TR Vios
jpeg_encoder	3h16m7s	73624	0
strVe_soc	3h14m0s	73271	0
aes256	1h35m51s	64435	0
genericfir	1h2m36s	48849	0
aes128	1h7m50s	44658	0
TEA	2h11m8s	44026	0
rc6_core	1h43m44s	35304	0
double_sqrt	1h14m18s	29252	0
iir5sfix	1h14m5s	24950	0
y_huff	0h54m48s	16826	0
sha3	0h21m18s	16372	0
ocs_blitter	0h17m48s	10997	0
sub86	0h12m35s	7655	0
CPU	0h11m7s	7342	0
cordic	0h10m13s	7210	0

Design for Testing (optional)

Scan insertion
Automatic test pattern generation
Test patterns compaction
Fault coverage
Fault simulation



PHYSICAL IMPLIMENTATION

Also called automated PnR (Place and Route)

M.SRIGHUKAN , 7th THE BVB , ERODE

CHIP DESIGNING COURSE

- Floor/power planning
- End coupling capacitors and tap cells insertion
- Placement: global and detailed
- Post placement optimization
- Clock tree synthesis(CTS)
- ROUTING : global and detailed

LEC (using yosys)

Everytime the netlist is modified,verified must be performed

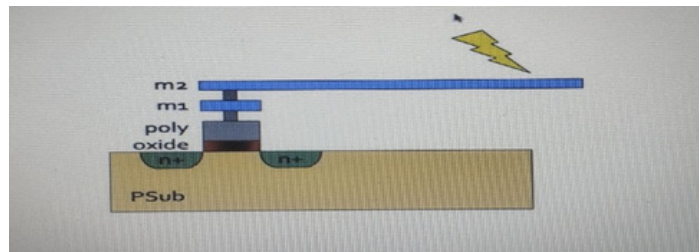
- CTS modifies the netlist
- Post placement optimizations modifies the net list

LEC is used to formally confirm that the function did not change after modifying the netlist

Dealing with antenna rules violations

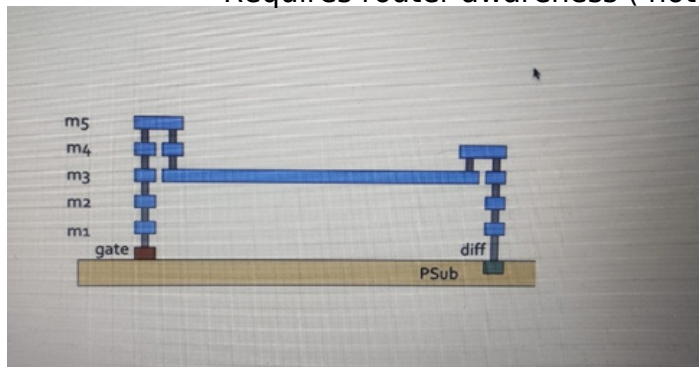
When a metal wire segment is frabricated ,it can act as an antenna

- Reactive ion etching causes charge to accumulate on the wire
- Transistor gates can be damaged during fabrication

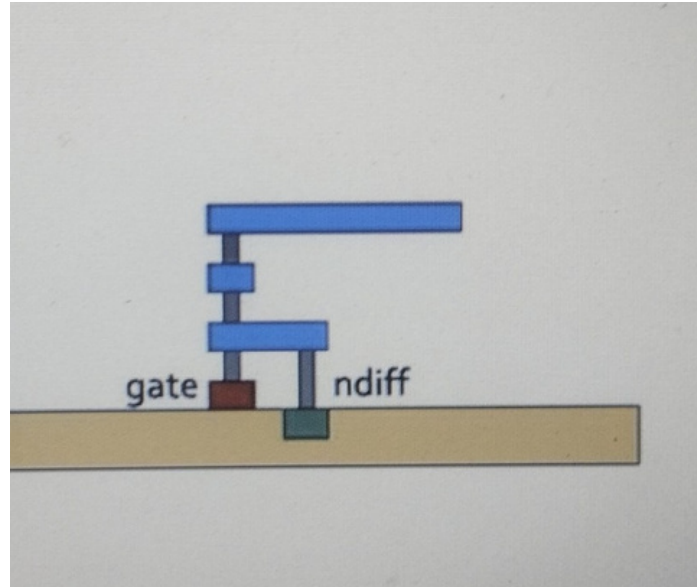


So the length of antennae should be limited. usually this is the job of the router . If the router fails to do so there are 2 solutions

- Bridging attaches a higher layer intermediary
Requires router awareness (not yet there)



- Add antenna diode cell to leak away charges
- Antenna diode are provided by the SCL



We took a preventive approach

- Add a fake antenna diode next to every cell input after placement
- Run the antenna checker (magic) on the routed layout
- If the checker reports a violation on the cell input pin, replace the fake diode cell by real one.

STATIC TIMING ANALYSIS

RC EXTRACTION : DEF2SPEF

STA: OpenSTA(OpenROAD)

Result gives violations if any

PHYSICAL VERIFICATION DRC & LVC

- Magic is used for design rules checking and SPICE extraction from layout
- Magic and netgen are used for LVS

Extracted SPICE by magic vs Verilog netlist.